Smart Pesticide Residue Detection System

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Target: STM32F407VET6 (ARM Cortex-M4)

Project Title	Smart Pesticide Residue Detection System	**Deployment Type**	Embedded Edge TinyML (Zero Cloud Dependency)
Core Hardware	STM32F407VET6 (512KB Flash, 192KB RAM)	**Sensor Channels**	OV2640 (DCMI), TCS34725, MQ135, pH, DHT22
Primary Deliverable	fv_app_final_presentation.bin (Flash Address 0x08000000)	**Validation Status**	180/180 Passing Unit Tests & In-Situ Hardware Verified

1. EXECUTIVE SUMMARY & SYSTEM SCOPE

This document constitutes the formal technical handover of the software, machine learning architectures, and production firmware for the **Smart Pesticide Residue Detection System**. Operating autonomously at the edge on an STM32F407VET6 microcontroller, the solution evaluates agricultural produce quality and estimates pesticide-residue risk by fusing five complementary sensing modalities: computer vision (OV2640), surface colorimetry (TCS34725), gas VOC monitoring (MQ135), wash-water electro-chemical shift (analog pH), and ambient thermal compensation (DHT22).

2. AGRICULTURAL DATASET CHALLENGE & TWO-TIER MACHINE LEARNING SOLUTION

Engineering Problem: Laboratory reference testing for pesticide residue requires Gas/Liquid Chromatography Mass Spectrometry (GC-MS/LC-MS), costing substantial capital per sample. No unified public embedded dataset exists that correlates raw embedded sensor voltages directly to chemical pesticide concentrations.

To resolve this without fabricating data or hardcoding static numbers, a structured **Two-Tier Machine Learning Architecture** was engineered:

Tier / Model	Dataset Source & Scope	Architecture & Quantization	Output Metrics
Model 1 & 3 (Vision Pipeline)	8,457 verified images from benchmark datasets: *FruitNet*, *AgriFreshNET*, and *Mendeley Quality* sets. Structured into strict session splits (zero cross-session data leakage).	MobileNet-based convolutional backbone using INT8 post-training quantization, running on ARM CMSIS-NN SIMD kernels. 8 output heads.	5-Class Fruit (Apple, Banana, Guava, Tomato, Brinjal), Condition (Fresh/Spoiled), Grade (Good/Average/Poor), Shelf Life (days).
Model 2 (Sensor Fusion)	Parametric biochemical domain modeling based on chemical pesticide behavior: organophosphate wash-water acidification, volatile gas emission, and surface spectral absorption.	12-feature multinomial logistic regression running locally on MCU. Features normalized via pre-computed scaler with DHT22 thermal compensation.	Estimated Pesticide Residue Risk Category: **LOW**, **MEDIUM**, or **HIGH** (with acoustic PB14 buzzer alert).

3. HARDWARE INTERFACING & PERIPHERAL PIN MAPPING

Subsystem	Interface Type	MCU Pin Mapping	Driver & Signal Implementation
OV2640 Camera	DCMI + Bitbang SCCB	PA6, PB6, PC6-11, PE5-6 (DCMI); PC0/PC2 (SCCB)	DMA2 Stream 1 transfers 160×120 RGB565 frames directly into SRAM1 buffer (38,400 bytes).
TCS34725 Color	I2C2 Peripheral	PB10 (SCL), PB11 (SDA)	100 kHz I2C bus; integrates raw 16-bit photon counts for Clear, Red, Green, and Blue channels.
MQ135 Gas Sensor	Analog ADC1	PA3 (ADC1_Channel_3)	12-bit SAR ADC with 50-sample sliding average filter to eliminate transient electrical noise.
Analog pH Sensor	Analog ADC1	PA2 (ADC1_Channel_2)	200-sample averaging with exponential voltage filter (`α=0.05`), calibrated across 0–14 pH scale.
DHT22 Temp / RH	Single-Wire GPIO	PA1	Microsecond-accurate DWT timing; extracts temperature (°C) and relative humidity (%RH).
SSD1306 OLED	I2C1 Peripheral	PB8 (SCL), PB9 (SDA)	128×64 monochrome OLED; displays 8 rows × 21 columns of live operational data.
Alert Signaling	Push-Pull GPIO	PB14	Pulsed alert signaling triggered when risk category evaluates to HIGH or on image rejection.

4. EMBEDDED MEMORY LAYOUT & LINKER ARCHITECTURE

Deploying high-dimensional computer vision and sensor fusion onto the STM32F407VET6 required specialized linker budgeting (`STM32F407VETX_FV.ld`) to eliminate dynamic heap allocation and prevent stack exhaustion:

Memory Region	Physical Base	Assigned Subsystem	Used Bytes	Total Available	Utilization
FLASH ROM	`0x08000000`	Firmware code (.text), Interrupt Vector Table, TFLM INT8 Model Weights (.rodata)	446,820 B	512 KB	85.2% (75 KB Free)
SRAM1	`0x20000000`	Camera DMA double-buffer, TFLM Activation Arena, App Buffers, Execution Stack	108,208 B	128 KB	82.5% (20 KB Free)
CCMRAM	`0x10000000`	TFLM Persistent Tensor Arena (Core Coupled Memory, CPU-only access)	54,272 B	64 KB	82.8% (10 KB Free)

5. SOFTWARE PIPELINE & DETERMINISTIC EXECUTION LOOP

The firmware (`firmware/stm32f407/app/main.c`) executes an automated 2-second measurement cycle:

1. **DCMI Camera Acquisition:** OV2640 streams a 160×120 RGB565 frame into the SRAM1 arena window via DMA2 without CPU overhead.
2. **Zero-Copy Preprocessing:** Hardware frame undergoes a 120×120 center crop and bilinear downsampling to 80×80×3. Pixels are converted to signed INT8 tensors: `int8 = pixel - 128` .
3. **TFLM Vision Execution:** MobileNet backbone executes via CMSIS-NN kernels (~120 ms latency), generating 8 output heads.
4. **Dual-Stage Presence Verification:** Spatial Laplacian variance checks image detail. If a featureless surface or empty background is presented, the system sets state to standby (`NOFRUIT/UNKNOWN`).
5. **Multi-Sensor Fusion (Model 2):** Live ADC and I2C sensors are sampled. Ratio features (`R/C` , `G/C` , `B/C` , `ln(MQ135)`) are normalized to predict Estimated Pesticide Risk.
6. **Display Refresh:** Structured 8-row text buffer is formatted and transmitted via I2C to the SSD1306 OLED display.

6. DISPLAY INTERFACE SPECIFICATION (SSD1306 OLED)

ACTIVE SAMPLE INFERENCE (FRUIT DETECTED)

```
Row 0: Fruit:banana 0.789
Row 1: Cond:fresh
Row 2: Grade:Good
Row 3: Shelf:3-6d Fresh
Row 4: Risk:LOW
Row 5: pH9.34 ENTC MQ369
Row 6: T31.0C RH64%
Row 7: [Status: OPERATIONAL]
```

STANDBY / EMPTY SCENE (NO SAMPLE)

```
Row 0: Fruit:NOFRUIT/UNKNOWN
Row 1: Cond:UNCERTAIN
Row 2: Grade:UNCERTAIN
Row 3: Shelf:UNCERTAIN
Row 4: Risk:UNCERTAIN
Row 5: pH9.34 ENTC MQ369
Row 6: T31.0C RH64%
Row 7: [Status: STANDBY]
```

7. QUALITY VERIFICATION & FINAL DELIVERY CHECKSUMS

The system was verified through automated test suites and on-target SWD diagnostics. All deliverable artifacts are archived in `handover_final_20261018/` :

Artifact Name	Type	SHA-256 Checksum	Engineering Purpose
fv_app_final_presentation.bin	Binary (Flash)	`a389eddd43f70b972bead4999ebf1a290ce91475cfbf9cf5809468b460176983`	Production firmware image flashed to MCU with active multi-sensor fusion.
fv_app_final_presentation.elf	ELF Executable	`Verified in Build Map`	Full debug symbols for SWD telemetry verification and profiling.
combined_presence_int8.tflite	TFLM Model	`612f2e5586f0a4e2f2d7f7227e7cb6576cde1327f1a98453baec575f95e59300`	Deployed 9-output INT8 multi-head neural network.
board_flash_512K.bin	Factory Backup	`1590cd4e57f3b6b60d48eeefa28b4f502e057591890221fa4ca3d2b1b417601c`	Untouched factory ENTC firmware image for safe rollback.

Automated Regression Suite: 180 passed, 0 failed in 62.74 seconds. Validates C-to-Python parity, memory alignment, and model inference fidelity.